

Methodological Notes for Firm-Level Return Processing and CARDI Predictability Tests

CARDI Project

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1 Purpose

This note documents the empirical workflow implemented in:

`Code/Firm_Test/Process_FirmReturns.`

The workflow constructs firm-level daily and monthly return datasets, estimates firm-level pure returns, aggregates HC/MC/LC pool-level returns and volatilities, constructs HC-versus-LC relative volatility measures, and tests whether lagged CARDI and related indicators predict future HC relative volatility.

The code is organized around `main.R`, with module behavior controlled by `config.R`. Helper functions are stored in:

`functions_load_data.R`, `functions_returns.R`, `functions_controls.R`,
`functions_pure_return.R`, `functions_pool_returns.R`, `functions_relative_volatility.R`,
`functions_predictability_CARDI.R`, `functions_utils.R`.

2 Data Inputs

The workflow combines the following data sources:

Input	Role
<code>Data/Processed/Input/HighCarbonIntens</code>	HC stock prices and market values
<code>Data/Processed/Input/MedCarbonIntens</code>	MC stock prices and market values
<code>Data/Processed/Input/LowCarbonIntens</code>	LC stock prices and market values
<code>Data/Processed/FamaFactors/FamaFactors_Daily.Rds</code>	Daily risk-free rate
<code>Data/Processed/FamaFactors/FamaFactors_Monthly.Rds</code>	Monthly risk-free rate
<code>Data/raw/Stocklist_industry.xlsx</code>	Industry classification
<code>Data/Processed/CARDI/Month_CARDI.xlsx</code>	Monthly CARDI measures
<code>Data/Processed/Macro/Month_Macro.csv</code>	Monthly macro controls
<code>Data/Processed/Data_FI_WIND.rds</code>	Firm financial controls
<code>Output/Carbon_Rank.rds</code>	Carbon emissions ranking
<code>Output/NewIndicators/Monthly/All_Indicators_Monthly.csv</code>	Monthly indicators
<code>Data/raw/Important_Carbon_Events.xlsx</code>	Carbon-related event controls

Each firm is assigned to a carbon group according to its source pool:

$$\text{CarbonType}_i \in \{\text{HC}, \text{MC}, \text{LC}\}.$$

3 Daily and Monthly Firm Returns

3.1 Daily Log Returns

Let $P_{i,d}$ denote the closing price of firm i on trading day d . The daily log return is:

$$R_{i,d} = \log P_{i,d} - \log P_{i,d-1}. \quad (1)$$

Let r_d^f denote the daily risk-free rate from the daily Fama-factor file. The daily excess return is:

$$XR_{i,d} = R_{i,d} - r_d^f. \quad (2)$$

3.2 Monthly Log Returns

Let $P_{i,m}^{\text{end}}$ denote the price on the last trading day of month m . The monthly log return is:

$$R_{i,m} = \log P_{i,m}^{\text{end}} - \log P_{i,m-1}^{\text{end}}. \quad (3)$$

Let r_m^f denote the monthly risk-free rate from the monthly Fama-factor file. The monthly excess return is:

$$XR_{i,m} = R_{i,m} - r_m^f. \quad (4)$$

Using log returns follows the standard asset-pricing convention that returns compound additively over time.

4 Annual Firm Controls

4.1 Control Variables

The workflow constructs annual firm controls from `Data_FI_WIND.rds` and related files. The core controls are:

$$X_{i,y} = (\log(\text{Asset}_{i,y}), \text{ROE}_{i,y}, \text{BM}_{i,y}, \text{CapExRatio}_{i,y}).$$

Book-to-market is computed as:

$$\text{BM}_{i,y} = \frac{\text{BookValue}_{i,y}}{\text{MarketValue}_{i,y-1}^{\text{last}}}, \quad (5)$$

where the denominator is the last observed market value from the previous year.

Capital expenditure intensity is:

$$\text{CapExRatio}_{i,y} = \frac{\text{CapitalExpenditure}_{i,y}}{\text{Asset}_{i,y}}. \quad (6)$$

The workflow also merges lagged log carbon emissions:

$$\log(\text{Emissions}_{i,y}),$$

but this variable is retained as an output field and is intentionally excluded from the pure-return regression.

4.2 Lagging Convention

For a return observation in calendar year y , the workflow uses previous-year controls:

$$X_{i,t}^{\text{lag}} = X_{i,y(t)-1}. \quad (7)$$

This lagging convention reduces look-ahead bias by ensuring that controls are measured before the return observation.

4.3 Winsorization

All control variables are winsorized at the 1st and 99th percentiles:

$$\tilde{X}_{i,y,k} = \min \{ \max (X_{i,y,k}, Q_{0.01,k}), Q_{0.99,k} \}. \quad (8)$$

Winsorization limits the influence of extreme accounting observations.

5 Pure Return Estimation

5.1 Motivation

The pure-return measure removes variation in firm excess returns associated with observable firm characteristics and industry fixed effects. The residual is interpreted as a return component net of these standard controls.

5.2 Regression Specification

For each frequency $f \in \{\text{daily}, \text{monthly}\}$, the workflow estimates a pooled OLS regression:

$$XR_{i,t}^{(f)} = \alpha + \beta' \tilde{X}_{i,t}^{\text{lag}} + \gamma_{g(i)} + \varepsilon_{i,t}^{(f)}, \quad (9)$$

where:

$$\tilde{X}_{i,t}^{\text{lag}} = (\text{lag_logAsset}_{i,t}, \text{lag_ROE}_{i,t}, \text{lag_BM}_{i,t}, \text{lag_CapExRatio}_{i,t}),$$

and $\gamma_{g(i)}$ is an industry fixed effect based on the firm industry code.

The residual is stored as:

$$\text{PureReturn}_{i,t}^{(f)} = \hat{\varepsilon}_{i,t}^{(f)}. \quad (10)$$

The emission variable `lag_logEmissions` is not included in the pure-return regression. This choice prevents the residualization step from mechanically removing carbon-related variation that may be relevant for the subsequent CARDI predictability tests.

6 Firm-Level Monthly Volatility

The monthly firm-level volatility measures are computed from daily observations within each firm-month. Let $D(i, m)$ be the set of trading days for firm i in month m . The return volatility is:

$$\sigma_{i,m}^R = \text{sd}(\{R_{i,d} : d \in D(i, m)\}). \quad (11)$$

The excess-return volatility is:

$$\sigma_{i,m}^{XR} = \text{sd}(\{XR_{i,d} : d \in D(i, m)\}). \quad (12)$$

The pure-return volatility is:

$$\sigma_{i,m}^{PR} = \text{sd}(\{\text{PureReturn}_{i,d} : d \in D(i, m)\}). \quad (13)$$

In the output datasets, these variables are named:

`vola_return`, `vola_ExReturn`, `vola_PurReturn`.

7 Pool-Level Returns and Volatility

7.1 Value-Weighted Pool Returns

For carbon group $c \in \{\text{HC}, \text{MC}, \text{LC}\}$, let $\mathcal{I}_{c,d}$ be the set of firms in group c with non-missing market value on day d . The value weight is:

$$w_{i,d}^c = \frac{\text{MktCap}_{i,d}}{\sum_{j \in \mathcal{I}_{c,d}} \text{MktCap}_{j,d}}. \quad (14)$$

The pool-level daily return is:

$$R_{c,d} = \sum_{i \in \mathcal{I}_{c,d}} w_{i,d}^c R_{i,d}. \quad (15)$$

Analogously, the workflow computes value-weighted pool excess returns and value-weighted pool pure returns:

$$XR_{c,d} = \sum_{i \in \mathcal{I}_{c,d}} w_{i,d}^c XR_{i,d}, \quad (16)$$

$$PR_{c,d} = \sum_{i \in \mathcal{I}_{c,d}} w_{i,d}^c \text{PureReturn}_{i,d}. \quad (17)$$

Value weighting follows the asset-pricing convention that larger firms receive larger portfolio weights, as in standard portfolio construction and factor work such as Fama and French (1993).

7.2 Pool-Level Monthly Volatility

Pool-level monthly volatility is computed from the daily pool-level series within each month:

$$\sigma_{c,m}^R = \text{sd}(\{R_{c,d} : d \in m\}), \quad (18)$$

$$\sigma_{c,m}^{XR} = \text{sd}(\{XR_{c,d} : d \in m\}), \quad (19)$$

$$\sigma_{c,m}^{PR} = \text{sd}(\{PR_{c,d} : d \in m\}). \quad (20)$$

7.3 Weighted Average Firm Volatility

The workflow also computes a value-weighted average of firm-level monthly volatility:

$$\bar{\sigma}_{c,m}^R = \sum_{i \in \mathcal{I}_{c,m}} w_{i,m}^c \sigma_{i,m}^R, \quad (21)$$

with analogous measures for excess-return and pure-return volatility.

8 HC-versus-LC Relative Volatility

The main outcome variables compare HC volatility to an LC benchmark.

8.1 Firm-Level Relative Volatility

For an HC firm i , relative volatility is:

$$\text{Rela_vola_Return}_{i,m} = \frac{\sigma_{i,m}^R}{\bar{\sigma}_{LC,m}^R}. \quad (22)$$

The analogous measures are:

$$\text{Rela_vola_ExReturn}_{i,m} = \frac{\sigma_{i,m}^{XR}}{\bar{\sigma}_{LC,m}^{XR}}, \quad (23)$$

$$\text{Rela_vola_PurReturn}_{i,m} = \frac{\sigma_{i,m}^{PR}}{\bar{\sigma}_{LC,m}^{PR}}. \quad (24)$$

8.2 Pool-Level Relative Volatility

For the HC pool:

$$\text{Rela_vola_Return}_{HC,m} = \frac{\bar{\sigma}_{HC,m}^R}{\bar{\sigma}_{LC,m}^R}. \quad (25)$$

The same construction is used for excess-return and pure-return volatility.

Relative volatility normalizes HC risk by the contemporaneous LC benchmark, which helps isolate the changing risk of high-carbon firms relative to low-carbon firms.

9 Predictability Regressions

9.1 Predictor Construction

The workflow tests whether lagged CARDI or alternative indicators predict future HC relative volatility. For a monthly predictor Z_m , the one-month lag is:

$$Z_{m-1}. \quad (26)$$

The main CARDI predictors include:

$$\text{CARDI_1P_M}, \quad \text{CARDI_5P_M}, \quad \text{CARDI_10P_M},$$

and their log-difference versions:

$$\text{CARDI_1P_LogDiff_M}, \quad \text{CARDI_5P_LogDiff_M}, \quad \text{CARDI_10P_LogDiff_M}.$$

9.2 Firm-Level Regressions

For HC firms, the generic panel predictability regression is:

$$Y_{i,m} = \alpha_i + \theta Z_{m-1} + \delta' M_m + \lambda' E_m + \psi' C_{i,m} + \rho T_m + \eta_{y(m)} + u_{i,m}, \quad (27)$$

where:

- $Y_{i,m}$ is one of the three relative volatility outcomes;
- Z_{m-1} is lagged CARDI or another indicator;
- M_m are macro controls;
- E_m are event controls;
- $C_{i,m}$ are firm controls;
- α_i are firm fixed effects;
- T_m is a year trend;
- $\eta_{y(m)}$ are year fixed effects.

The implemented firm-level specifications are:

$$\begin{aligned} \text{NoControls :} \quad & Y_{i,m} = \alpha + \theta Z_{m-1} + u_{i,m}, \\ \text{ControlsFirmFE :} \quad & Y_{i,m} = \alpha_i + \theta Z_{m-1} + \delta' M_m + \lambda' E_m + \psi' C_{i,m} + u_{i,m}, \\ \text{ControlsFirmFEYearTrend :} \quad & Y_{i,m} = \alpha_i + \theta Z_{m-1} + \delta' M_m + \lambda' E_m + \psi' C_{i,m} + \rho T_m + u_{i,m}, \\ \text{ControlsFirmFEYearFE :} \quad & Y_{i,m} = \alpha_i + \theta Z_{m-1} + \delta' M_m + \lambda' E_m + \psi' C_{i,m} + \eta_{y(m)} + u_{i,m}. \end{aligned}$$

Firm-level standard errors are clustered by firm when clustering is enabled. This follows the finance-panel inference concerns discussed by Petersen (2009).

9.3 Pool-Level Regressions

For HC pool-level data, the generic regression is:

$$Y_m = \alpha + \theta Z_{m-1} + \delta' M_m + \lambda' E_m + \rho T_m + \eta_{y(m)} + u_m. \quad (28)$$

Pool-level regressions use Newey–West standard errors to account for heteroskedasticity and serial correlation in monthly time-series residuals (Newey and West, 1987).

9.4 Inference

For time-series models, the Newey–West covariance estimator is:

$$\widehat{\text{Var}}_{\text{NW}}(\hat{\beta}) = (X'X)^{-1} \left(\sum_{\ell=-L}^L w_\ell \sum_{t=|\ell|+1}^T x_t x'_{t-|\ell|} \hat{u}_t \hat{u}'_{t-|\ell|} \right) (X'X)^{-1}, \quad (29)$$

where $w_\ell = 1 - |\ell|/(L+1)$. The configured lag is:

$$L = 12.$$

For firm panels, the clustered covariance estimator allows arbitrary residual correlation within firm:

$$\widehat{\text{Var}}_{\text{CL}}(\hat{\beta}) = (X'X)^{-1} \left(\sum_{i=1}^N X_i' \hat{u}_i \hat{u}_i' X_i \right) (X'X)^{-1}, \quad (30)$$

with finite-sample adjustment handled by the implementation.

10 Interpretation of the CARDI Coefficient

The central coefficient is θ . A positive estimate implies:

$$\theta > 0 \quad \Rightarrow \quad \text{higher lagged CARDI predicts higher future HC relative volatility.}$$

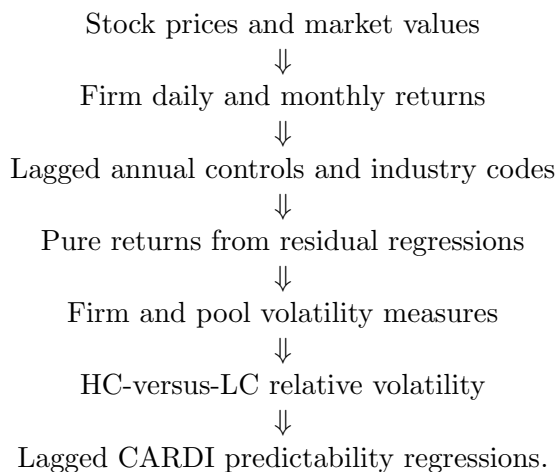
This is consistent with the idea that aggregate carbon-risk dynamics contain information about future volatility pressure among high-carbon firms. The economic motivation is related to equilibrium sustainable investing models, in which ESG preferences affect asset demand, expected returns, realized returns around shocks, and firms' costs of capital (Pástor, Stambaugh, and Taylor, 2021).

11 Output Files

Output	Description
<code>Firm_return_daily.rds</code>	Firm-level daily returns, excess returns, controls, pure returns
<code>Firm_return_month.rds</code>	Firm-level monthly returns and monthly volatility
<code>HC_daily.rds</code> , <code>MC_daily.rds</code> , <code>LC_daily.rds</code>	Pool daily returns
<code>HC_monthly.rds</code> , <code>MC_monthly.rds</code> , <code>LC_monthly.rds</code>	Pool monthly volatility and enrichment variables
<code>HC_Firm_Monthly.rds</code>	HC firm-level relative volatility dataset
<code>HC_Monthly.rds</code>	HC pool-level relative volatility dataset
<code>workflow_validation_summary.csv</code>	Observation counts and validation checks
<code>Predictability/*.csv</code>	Predictor-specific and model-specific regression outputs

12 Workflow Summary

The complete workflow is:



References

References

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